DigiLock 110 Documentation

Hardware-Version: 1V3

Module Overview

Application: Laser Stabilization on Cavity (PDH)

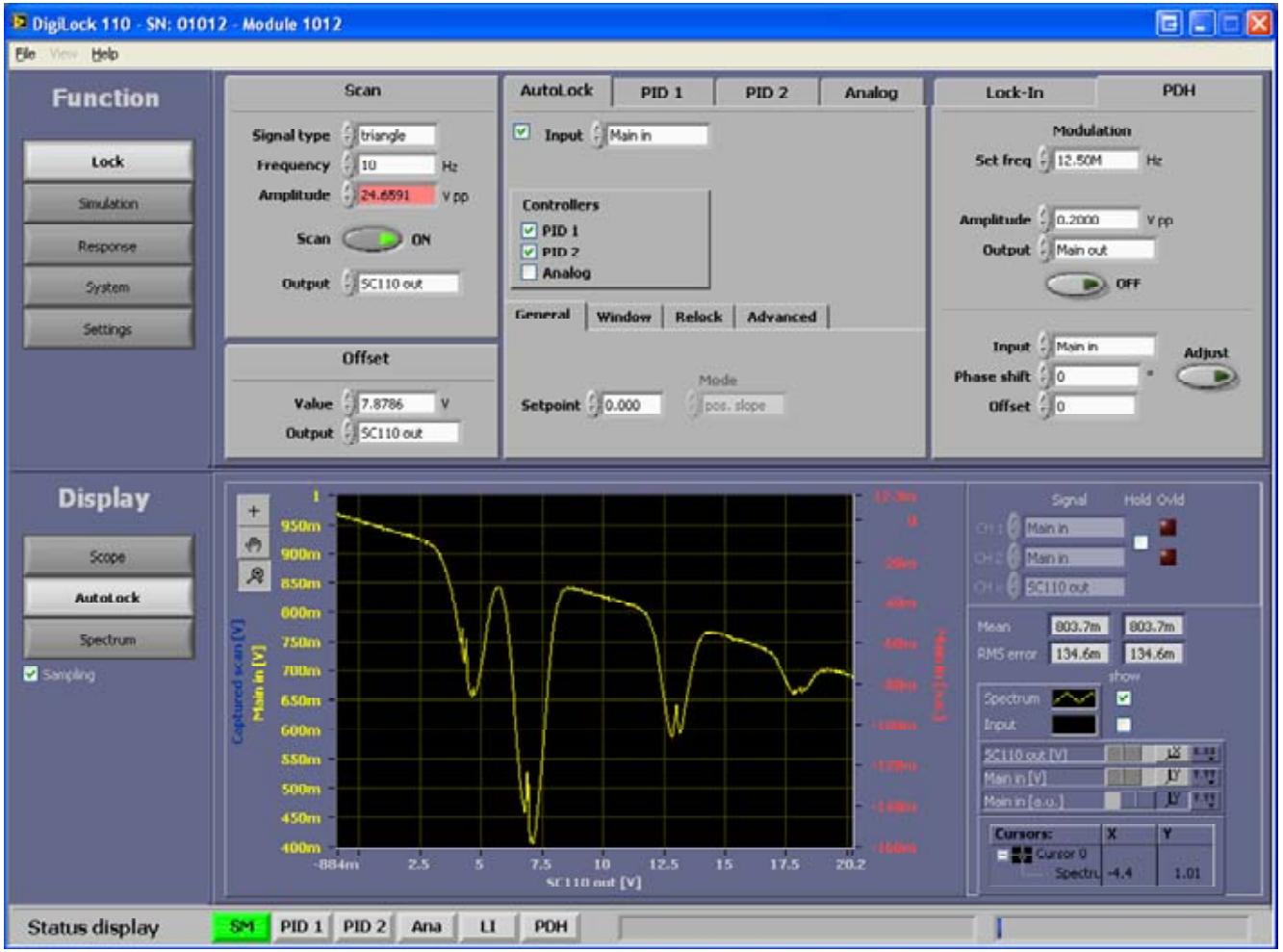
Application: Laser Stabilization on Spectroscopy

Electrical Specifications

User Interface Software

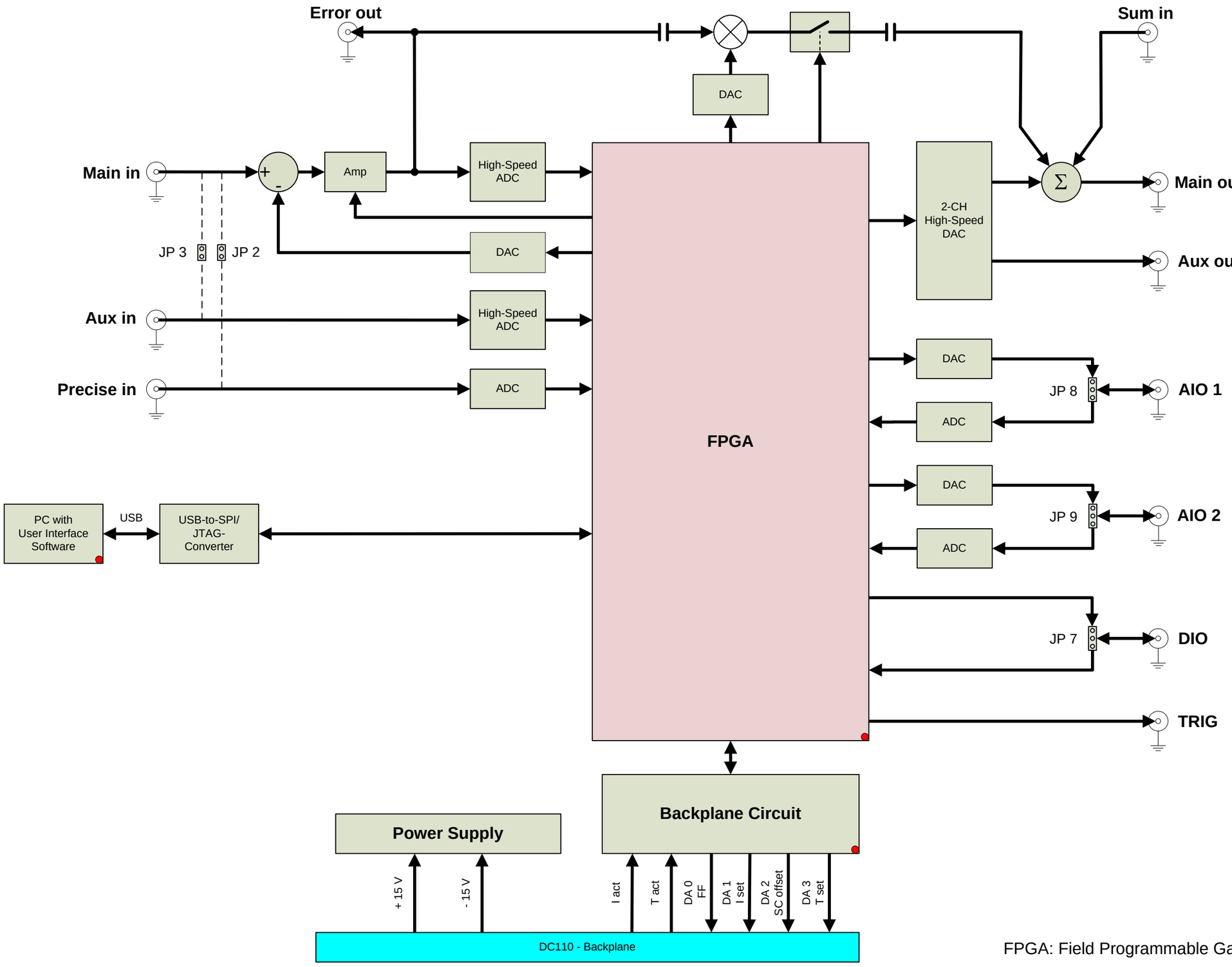
Module Overview

- Graphical user interface to control the DigiLock module
- Consists of „virtual modules“: Scan Module, 2-channel PID, Lock-in-/PDH-Module
- Digital 2-channel oscilloscope (in xy-mode: 3 channels are available)
- Support of interactive locking mechanisms (Side-of-Fringe, Top-of-Fringe)
- FFT-Spectrum Analyzer to optimize the control loop
- Network Analyzer to determine the transfer function of the system



Module Overview

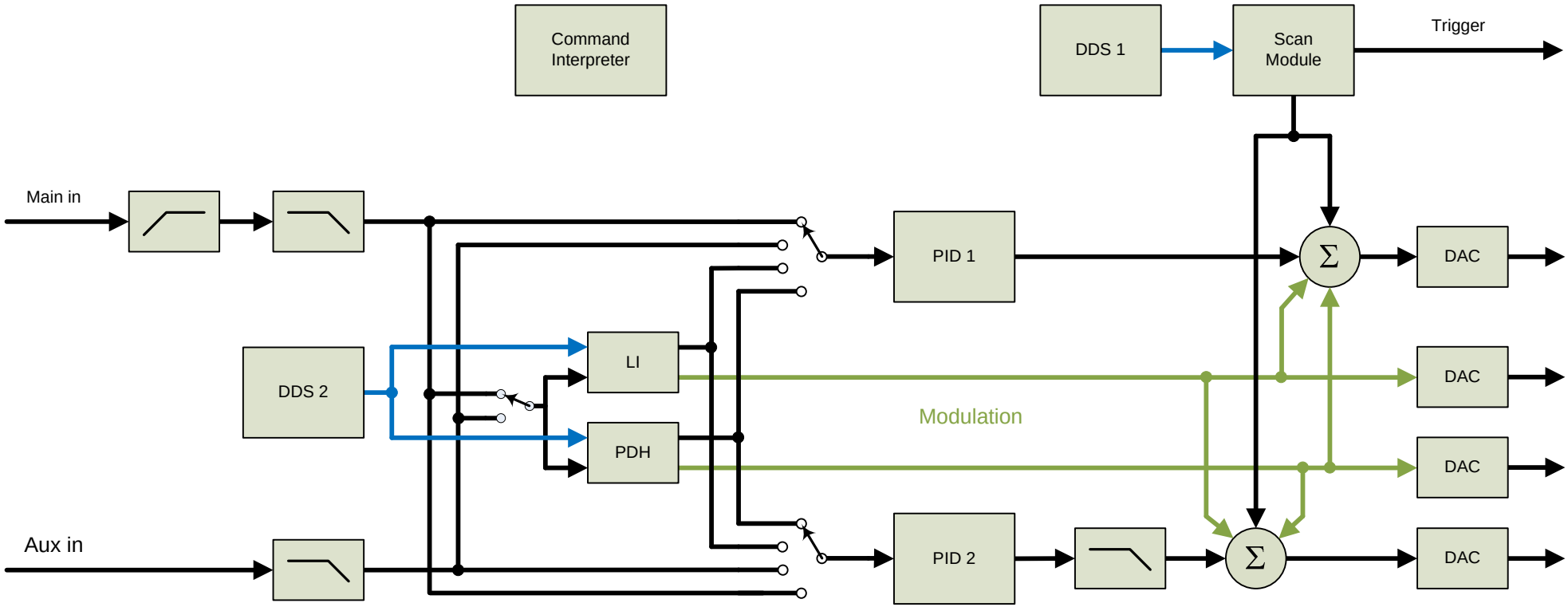
Main



FPGA: Field Programmable Gate Array

FPGA Module

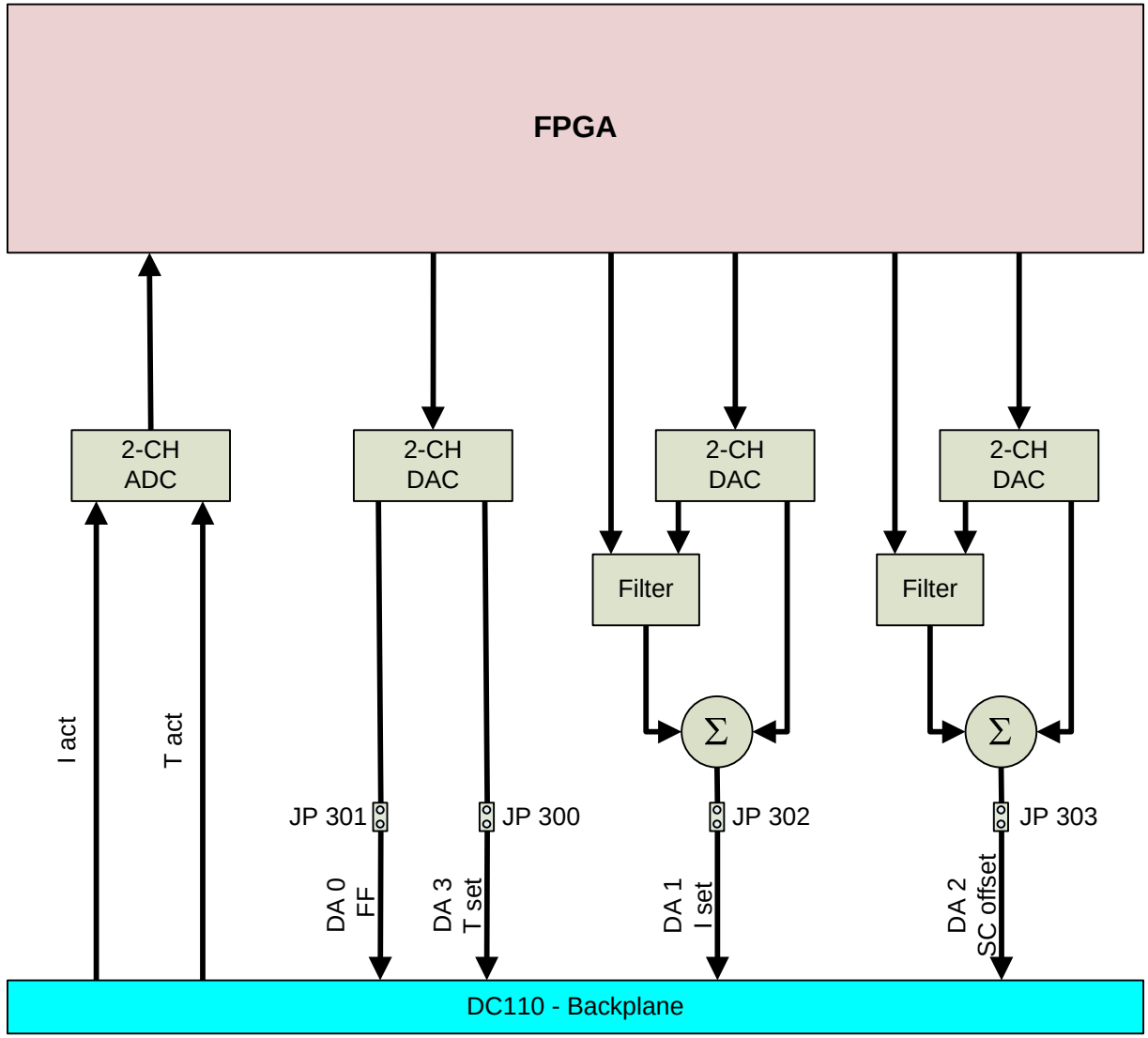
Module Overview



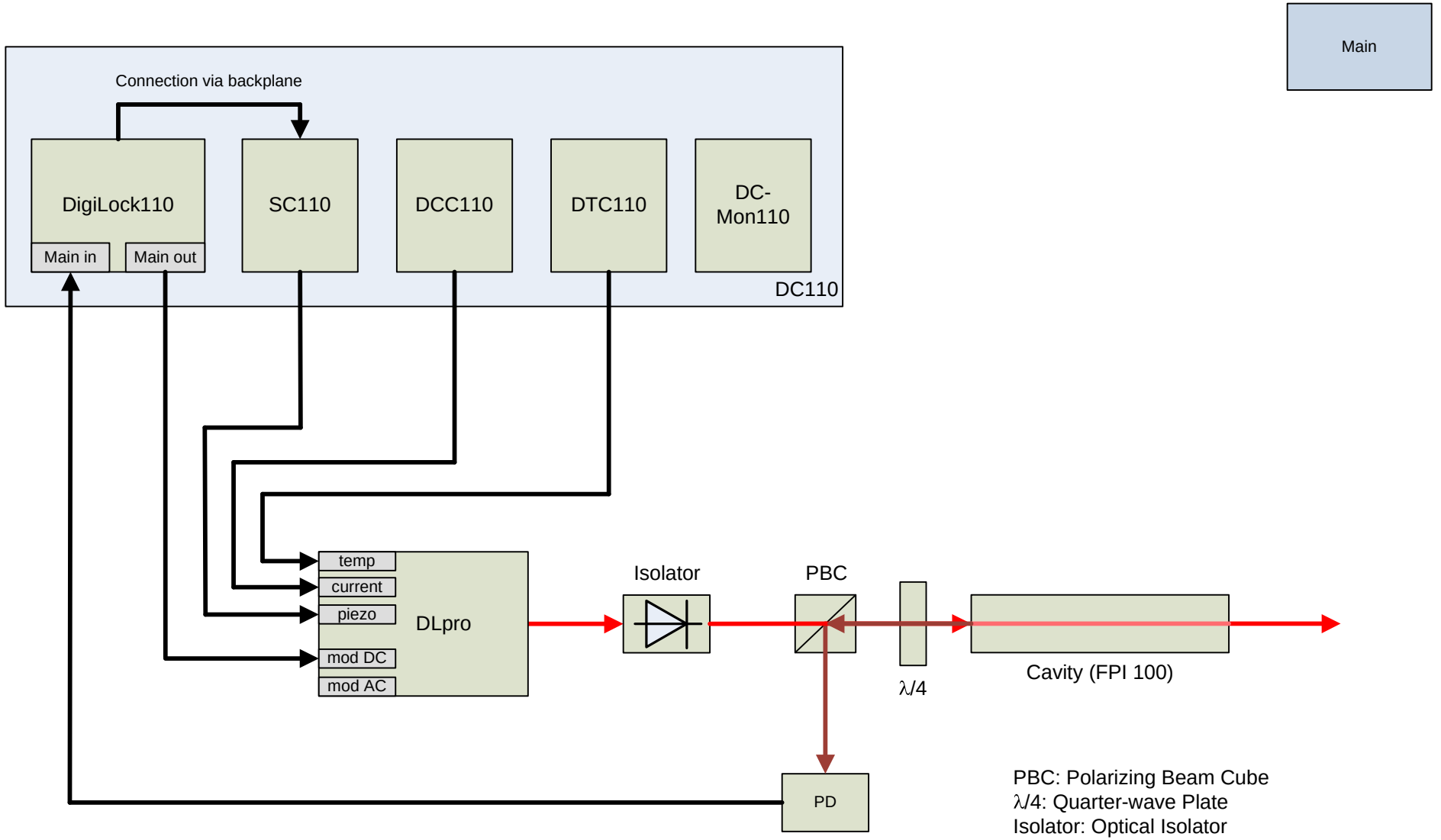
DDS: Direkt Digital Synthesizer
LI: Lock-in
PDH: Pound-Drever-Hall
ADC: Analog-to-Digital Converter
DAC: Digital-to-Analog Converter
FPGA: Field Programmable Gate Array

Backplane Circuit

Module Overview

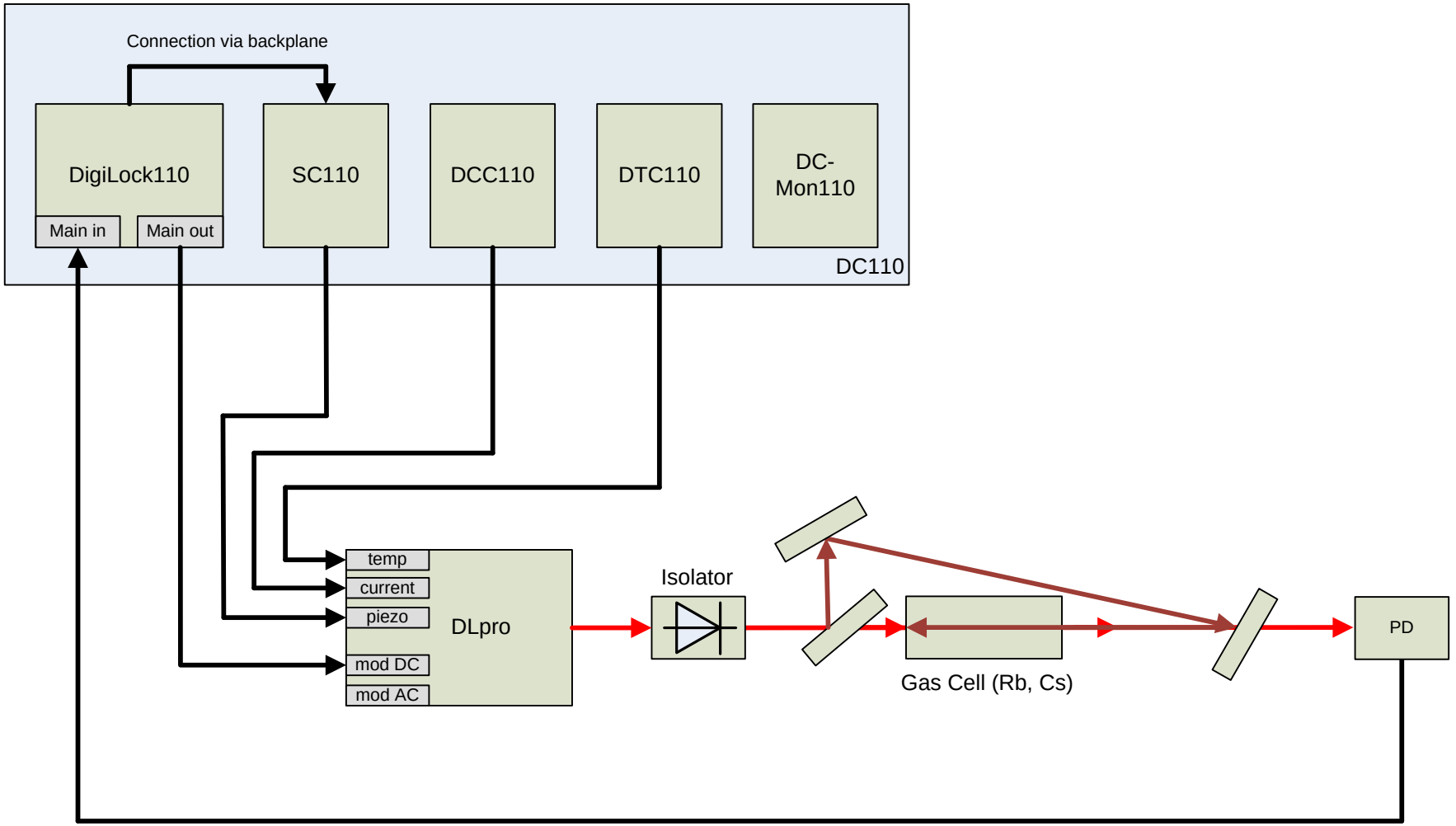


Application: Laser Stabilization on Cavity (PDH)



Application: Laser Stabilization on Doppler-free Saturation Spectroscopy

Main



Electrical Specifications

Hardware-Version: 1V3

Main

Power Supply

	Value
Supply Voltage	+-15 V
Supply Current +15V (max)	700 mA
Supply Current -15V (max)	200 mA

Input Channel	Resolution [bit]	Sample Rate [Hz]	Bandwidth (- 3dB) [Hz]	Range [V]	Input impedance [Ohm]	Comment
Main in	14	100 M	14 M	+- 2.0	50	input signal at <Main in> has to be between +- 3.5V; it can be shifted with the <Input Offset> and amplified with the input gain
Aux in	14	100 M	15 M	+- 2.0	50	
Precise in	16	200 k	50 k	+- 2.0	10 k	
DCC Iact	16	100 k	15 k	+- 13.1	40 k	backplane; default: not connected
DTC Tact	16	100 k	15 k	+- 13.1	> 500 k (40 k in Version 1V1)	backplane; default: not connected
AIO 1 in	16	100 k	15 k	+- 12.5	47 k	default: not connected
AIO 2 in	16	100 k	15 k	+- 12.5	47 k	
Sum in			27 M	+- 1.0	50	bandwidth value refers to signal path between <Sum in> and <Main out>
DIO in	1			0, 2.6	47 k	default: not connected; TTL-Level

Output Channel	Resolution [bit]	Sample Rate [Hz]	Bandwidth (- 3dB) [Hz]	Range [V]	50 Ohm Driver	Comment
Main out	14	100 M	19 M	+- 1.0	Yes	sum of <Sum in> and Analog P branch
Aux out	14	100 M	19 M	+- 1.0	Yes	
SC110 out	21	100 k	18 k	+- 6.5	No	backplane; amplification by 15 with SC110
DCC Iset	21	100 k	18 k	+- 6.5	No	backplane
DTC Tset	16	100 k	18 k	+- 6.5	No	backplane
AIO 1 out	16	100 k	16 k	+- 6.5	No	
AIO 2 out	16	100 k	16 k	+- 6.5	No	default: not connected
Error out			20 M	+- 1.7	Yes	Error out = (<Main in> + <Input Offset>) * Gain / 2; bandwidth value refers to signal path between <Main in> and <Error out>
TRIG	1			0, 2.6	Yes	TTL-Level
DIO out	1			0, 2.6	Yes	TTL-Level

Internal Signal	Resolution [bit]	Sample Rate [Hz]	Bandwidth (- 3dB) [Hz]	Range [V]		Comment
Input Offset	16	100 k	-	+- 2.5	-	
Analog P	16	100 k	-	+- 0.58	-	